Supplementary Material: Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Assessment of Heterogeneous Second-Life EV Batteries

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This supplement gives the preprocessing, the input correlations, the training curves and the full classification and regression metrics behind the paper. Every figure and table is generated by a script in the repository from committed results or from the public data; the script is named in each caption. SOH is relative to each cell's first valid capacity check.

S1. Preprocessing

Preprocessing is the same for every model and uses no label information beyond the SOH history at the origin.

1) *Parsing and verification.* Raw files are verified against their repository record or checksums. Our parsing reproduces the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks exactly.
2) *Validity rules.* An S6 check is used only if its C/25 discharge starts from full charge and reaches the cut-off; 4 of 847 checks fail (Fig. S1c). In S1, RPT 2 (run at about 20 °C) is dropped.
3) *Noise reduction.* Incremental-capacity curves $-\mathrm{d}Q/\mathrm{d}V$ are computed on a 1 mV (S6) or 2 mV (S1) voltage grid after a Gaussian filter of the charge (5 mV in S6, 10 mV in S1); without it, differentiation amplifies the measurement noise (Fig. S1b). Peak positions and heights are taken from the smoothed curve; in S6 a peak at the edge of its window is set to missing. This is the only filtering; no other denoising or outlier removal is applied.
4) *Ageing cycles.* Apache Spark reduces 9.0×10^8 per-second rows of the S6 ageing cycles to block inputs (mean, last value and slope of seven responses per 100-cycle block).
5) *Inputs and scaling.* Features never observed are dropped (e.g. the second IC window in S6). Missing values are imputed with the training median and flagged; inputs are standardised. Imputation, scaling and target scaling are fitted on the first task and then frozen.

S2. Input correlations

Fig. S3 gives the Spearman correlation of the S6 history and diagnostic inputs (F1) with each other and with the target. Ten of the 17 diagnostics are near-monotone functions of the SOH at the origin ($|\rho| \geq 0.9$), so they add little beyond the SOH history; the fade rate is the input most correlated with the target ($\rho = 0.8$). This is consistent with the ablation in the paper, where removing the diagnostics improved the TFT.

S3. Training curves

Each task is trained with Adam (batches of 64) for at most 200 epochs, with early stopping on the task's validation RMSE (patience 20); the weights of the best epoch are kept. Fig. S4 shows one stream of the residual TFT + EWC (horizons 1–3, first evaluation order, seed 0, tuned learning rate $3 \cdot 10^{-3}$ and $\lambda = 1$). The re-run reproduces the committed final average error of that stream exactly (0.464 pp). Validation RMSE reaches its minimum within 2–30 epochs on five tasks and after 86 epochs on task 6; on task 5 no epoch improved on the weights carried over from task 4, which were therefore kept.

S4. Full metric sets (P2)

Tables SI and SII give every regression and classification metric of the P2 forecasts (decision D30); Fig. S5 gives the S6 confusion matrices and Fig. S6 the error per held-out group.

S5. Additional results

Table SIII and Figs. S7 and S8 belong to Sections IV-E to IV-H of the paper and were moved here for space.

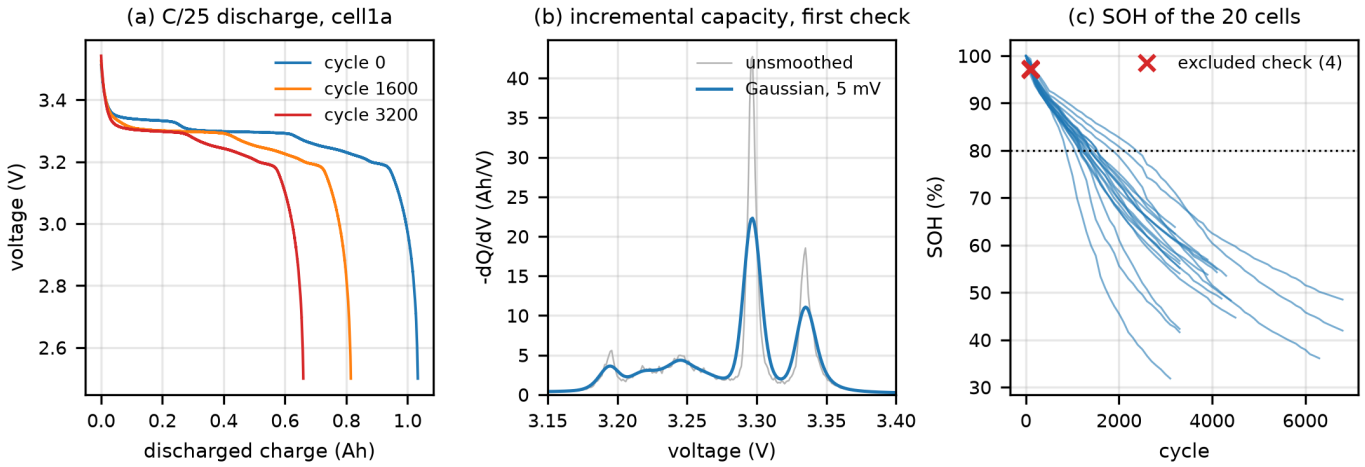


Fig. S1. S6 preprocessing (scripts/make_preprocessing_figures.py). (a) C/25 discharge of cell 1a at three ages. (b) Incremental capacity of its first check without and with the 5 mV Gaussian smoothing used for the IC features. (c) SOH of the 20 cells at every valid check; crosses: the four checks excluded by the validity rule, placed on the cell's curve at the cycle they were taken (they have no valid capacity). Dotted: 80% SOH.

TABLE SI

PROTOCOL P2, ALL REGRESSION METRICS OF THE FORECAST SOH (PP UNLESS STATED), MEAN OVER HELD-OUT GROUPS (THE RMSE COLUMN EQUALS TABLE P2 OF THE PAPER FOR THE BASELINES; MLP AND TFT ARE RE-RUNS ON ANOTHER PLATFORM, WITHIN 0.0052 PP OF IT). BIAS = MEAN(FORECAST − MEASURED): POSITIVE VALUES OVERESTIMATE SOH. NRMSE = RMSE / RANGE OF MEASURED SOH IN THE GROUP. R^2 IS ON THE SOH LEVEL, R^2_Δ ON THE SOH CHANGE SINCE THE ORIGIN (PERSISTENCE HAS $R^2_\Delta \leq 0$). SOH IS RELATIVE TO THE FIRST OBSERVED CHECK.

Data	Method	RMSE	MSE	MAE	MedAE	Bias	NRMSE	MAPE (%)	R^2	R^2_Δ
S1	Persistence	0.545	0.303	0.470	0.473	+0.430	0.161	0.481	0.672	-1.853
S1	Linear extrapolation	0.511	0.293	0.365	0.226	-0.014	0.148	0.374	0.690	-1.314
S1	Ridge, usage only	0.301	0.096	0.206	0.133	-0.038	0.088	0.212	0.897	0.181
S1	Ridge, F0	0.418	0.212	0.369	0.350	+0.205	0.118	0.379	0.798	-0.673
S1	Ridge, F1	0.320	0.115	0.266	0.241	+0.065	0.092	0.273	0.881	0.083
S1	Gradient boosting, F1	0.219	0.053	0.148	0.102	-0.021	0.064	0.152	0.944	0.543
S1	MLP	0.285	0.087	0.211	0.162	+0.009	0.083	0.216	0.907	0.260
S1	TFT	0.291	0.090	0.210	0.149	+0.027	0.085	0.215	0.904	0.216
S6	Persistence	2.817	8.389	2.451	2.225	+2.451	0.056	3.539	0.956	-3.160
S6	Linear extrapolation	1.372	1.946	0.738	0.333	+0.050	0.028	0.973	0.988	-0.050
S6	Ridge, usage only	1.177	1.607	0.864	0.671	-0.003	0.023	1.292	0.993	0.275
S6	Ridge, F0	0.859	0.905	0.613	0.439	+0.028	0.016	0.893	0.996	0.614
S6	Ridge, F1	1.137	1.767	0.749	0.550	+0.013	0.022	1.130	0.991	0.221
S6	Gradient boosting, F1	0.710	0.725	0.488	0.321	+0.097	0.013	0.710	0.997	0.736
S6	MLP	0.826	1.018	0.608	0.444	+0.220	0.015	0.914	0.996	0.656
S6	TFT	0.868	1.151	0.621	0.424	+0.235	0.016	0.928	0.996	0.622

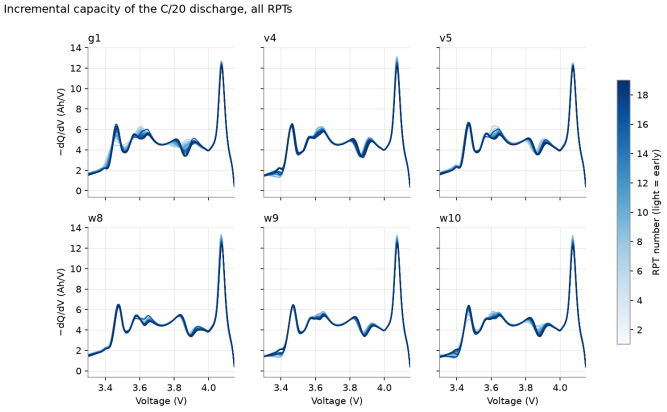


Fig. S2. S1: incremental-capacity curves of the C/20 RPTs (10 mV smoothing), notebook 11.

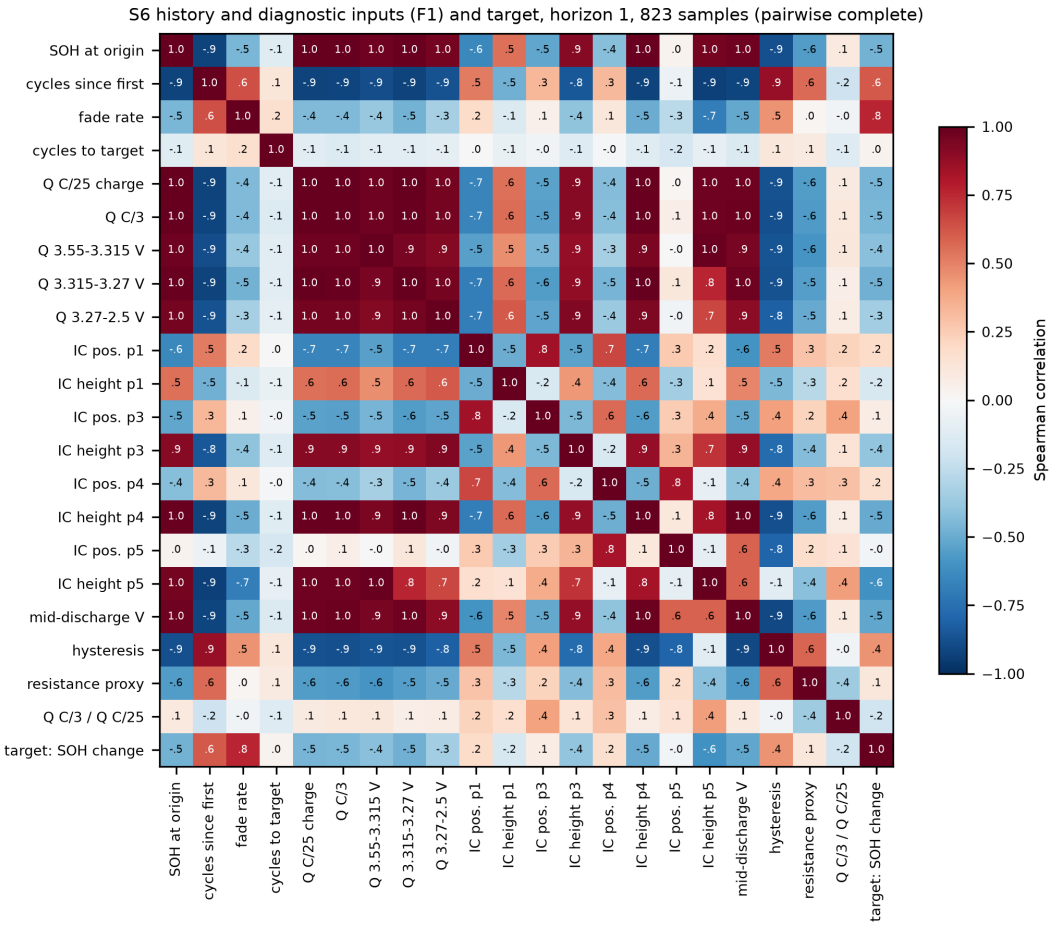


Fig. S3. Spearman correlation of the S6 F1 inputs and the target Δ (horizon 1, pairwise complete samples; the static test conditions are omitted). scripts/make_preprocessing_figures.py.

TABLE SII

PROTOCOL P2, REUSE CLASS (C0–C3, D10 CUT-POINTS) AND END-OF-FIRST-LIFE DECISION (SOH $\leq$ 80, POSITIVE) OF THE FORECAST SOH AGAINST THE MEASURED SOH, ALL HELD-OUT SAMPLES POOLED (MEAN OVER SEEDS). MACRO AVERAGES RUN OVER THE MEASURED CLASSES. AUC USES THE FORECAST SOH AS THE SCORE (POINT FORECASTS; NO CLASS PROBABILITIES, SO NO LOG LOSS). MEASURED CLASS COUNTS OF ALL SAMPLES: S1: C0 279, C1 0, C2 0, C3 0; S6: C0 52, C1 182, C2 516, C3 1659. – : UNDEFINED (ONE CLASS ONLY).

Data	Method	Acc.	Bal. acc.	Prec.	Rec.	Spec.	F1	MCC	κ	κ_w	EoL Prec.	EoL Rec.	EoL Spec.	EoL F1	EoL MCC	EoL AUC
S1	Persistence	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Linear extrapolation	0.98	0.98	1.00	0.98	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, usage only	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, F0	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Ridge, F1	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Gradient boosting, F1	0.99	0.99	1.00	0.99	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	MLP	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	TFT	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S6	Persistence	0.86	0.78	0.62	0.78	0.96	0.66	0.72	0.72	0.81	1.00	0.93	1.00	0.96	0.89	1.00
S6	Linear extrapolation	0.94	0.84	0.81	0.84	0.98	0.78	0.88	0.87	0.91	1.00	0.98	0.99	0.99	0.96	1.00
S6	Ridge, usage only	0.95	0.91	0.83	0.91	0.98	0.85	0.90	0.90	0.93	0.99	0.98	0.98	0.99	0.96	1.00
S6	Ridge, F0	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	1.00	0.98	0.99	0.99	0.97	1.00
S6	Ridge, F1	0.96	0.84	0.88	0.84	0.98	0.85	0.91	0.91	0.93	0.99	0.99	0.97	0.99	0.96	1.00
S6	Gradient boosting, F1	0.97	0.89	0.89	0.89	0.99	0.89	0.93	0.93	0.95	0.99	0.99	0.98	0.99	0.97	1.00
S6	MLP	0.96	0.86	0.88	0.86	0.99	0.87	0.92	0.92	0.94	0.99	0.99	0.98	0.99	0.97	1.00
S6	TFT	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	0.99	0.98	0.99	0.99	0.96	1.00

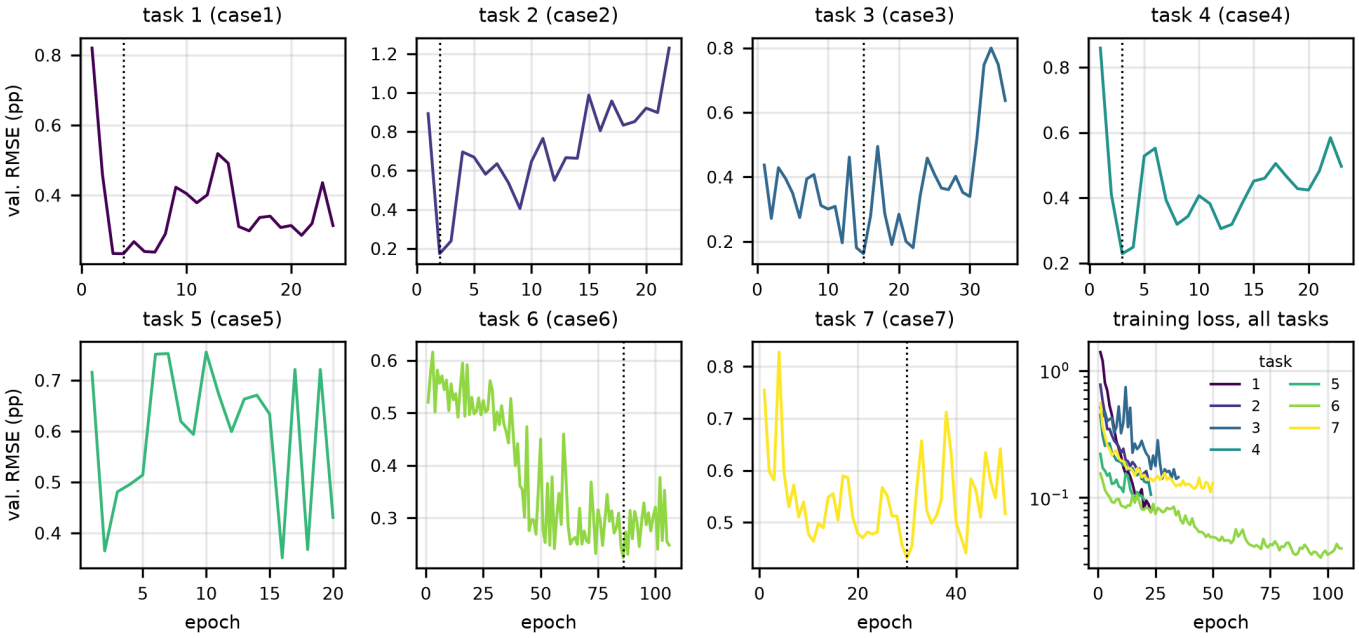


Fig. S4. Residual TFT + EWC on the S6 stream: validation RMSE per epoch for each task (dotted: early-stopping epoch) and training loss (scaled MSE plus EWC penalty) of all tasks. scripts/run_epoch_curves.py, results/epoch_curves_d23_e1.json.

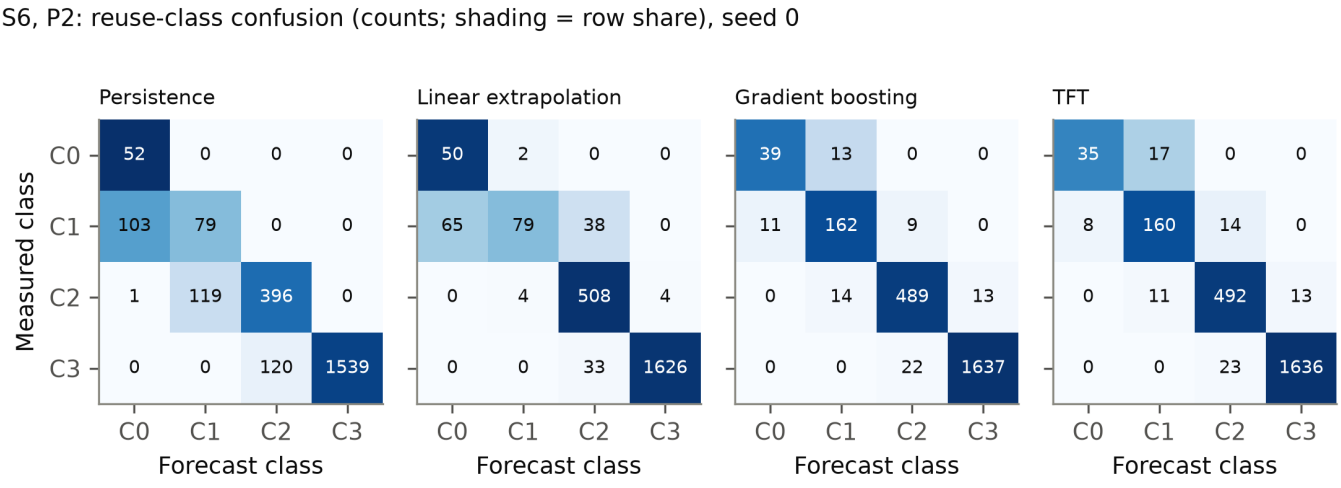


Fig. S5. S6, P2: reuse-class confusion matrices (counts; shading: row share), seed 0. scripts/make_d30_outputs.py.

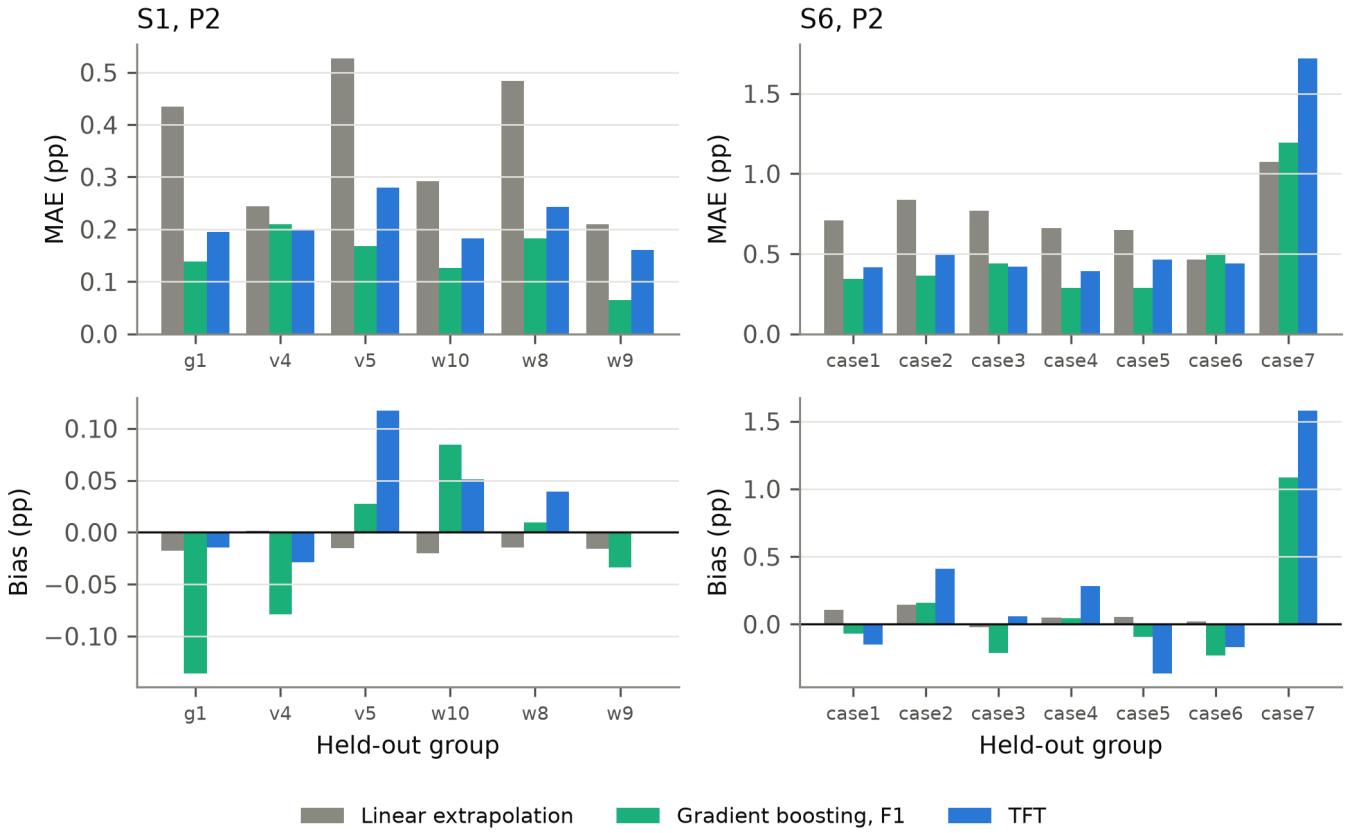


Fig. S6. P2: error per held-out group. scripts/make_d30_outputs.py.

TABLE SIII

LEARNED RESIDUAL, SHIFT-ADAPTIVE EWC AND LITERATURE BASELINES ON S6 (STREAM FROM SCRATCH): FINAL AVERAGE TEST RMSE (PP), MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS, AND FORGETTING (PP) FOR HORIZONS 1–6. BELOW THE LINE: SINGLE FITS ON ALL TASKS (NOT CONTINUAL) AND EXTRAPOLATION.

Method	E1: h=1–3	E2: h=1–6	Forgetting, E2
TFT + EWC, residual	**0.339 / 0.314**	**0.561 / 0.534**	**0.093 / 0.086**
TFT + SA-EWC, residual	0.343 / 0.321	0.548 / 0.564	0.066 / 0.084
TFT fine-tuning, residual	0.347 / 0.337	0.673 / 0.612	0.282 / 0.220
TFT + EWC, direct	0.504 / 0.487	0.802 / 0.835	0.135 / 0.166
MLP + EWC, residual	0.374 / 0.366	0.659 / 0.694	0.205 / 0.225
BiLSTM + EWC, residual	0.328 / 0.370	0.669 / 0.765	0.229 / 0.318
Transformer + EWC, residual	0.360 / 0.366	0.609 / 0.705	0.129 / 0.198
TCN + EWC, residual	0.466 / 0.408	0.812 / 0.756	0.345 / 0.268
Linear extrapolation	0.343	0.485	–
Gradient boosting	0.451	0.898	–
SVR, residual	0.429	0.616	–
GPR, direct	1.374	2.073	–

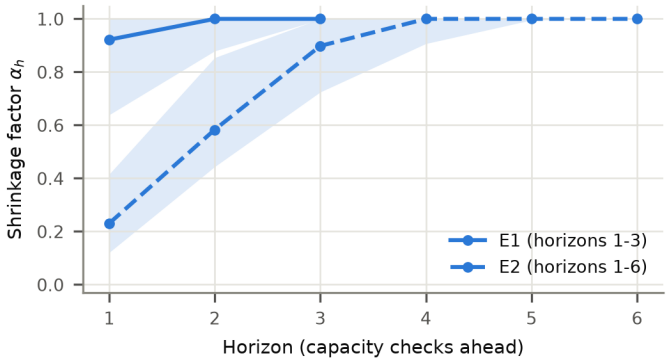


Fig. S7. S6 stream: shrinkage factor α_h of the TFT correction per horizon after the last task (median and interquartile range over 50 runs, seeds 0–4); $\alpha_h = 0$ is linear extrapolation.

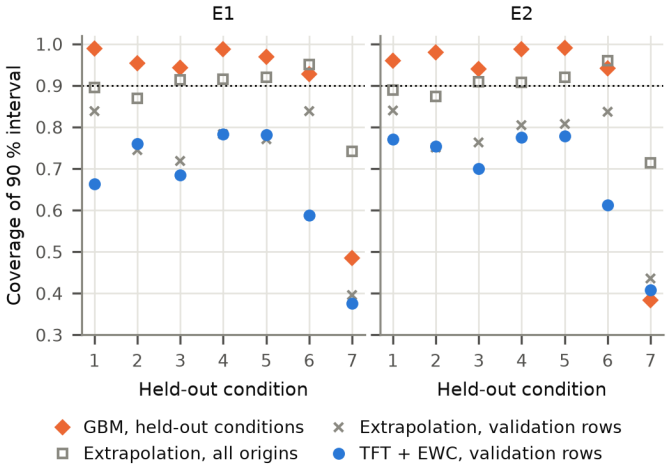


Fig. S8. Unseen S6 conditions: coverage of 90% conformal intervals per held-out condition, by forecaster and calibration rows. TFT: mean over 50 runs (seeds 0–4). Dotted: nominal level.